

Programming Fundamentals – Group Task

Assignment 01 – C++ Solutions

Group Members

Faiza Bibi — Roll No. 7

Noor Ul Ain — Roll No. 17

Syeda Arzoo — Roll No. 30

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These solutions are written in a simple student style with short explanations.

Q1. Sales Revenue Calculator

Take product name, quantity and selling price as input and find total revenue.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string productName;
    int quantity;
    double sellingPrice, revenue;

    cout << "Enter product name: ";
    getline(cin, productName);
    cout << "Enter quantity sold: ";
    cin >> quantity;
    cout << "Enter selling price: ";
    cin >> sellingPrice;

    revenue = quantity * sellingPrice;

    cout << "Product: " << productName << endl;
    cout << "Quantity Sold: " << quantity << endl;
    cout << "Total Revenue: Rs. " << revenue << endl;
    return 0;
}
```

Simple explanation: I used quantity × selling price to get the total revenue.

Q2. Student Business Profile

Take the student's name, age, semester, CGPA and whether the student is in Business Analytics.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    int age, semester;
    double cgpa;
    bool isAnalytics;

    cout << "Enter name: ";
    getline(cin, name);
    cout << "Enter age: ";
    cin >> age;
    cout << "Enter semester: ";
    cin >> semester;
    cout << "Enter CGPA: ";
    cin >> cgpa;
    cout << "Business Analytics student? (1/0): ";
    cin >> isAnalytics;

    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Semester: " << semester << endl;
    cout << "CGPA: " << cgpa << endl;
    cout << "Business Analytics Student: "
        << (isAnalytics ? "Yes" : "No") << endl;
}
```

```

    return 0;
}

```

Simple explanation: Different data types are used for name, age, CGPA and the yes/no value.

Q3. Business Profit Calculator

Find total revenue, total cost and profit using units sold, selling price, cost per unit and fixed cost.

```

#include <iostream>
using namespace std;

int main() {
    int units;
    double sellingPrice, costPerUnit, fixedCost;
    double revenue, variableCost, totalCost, profit;

    cin >> units >> sellingPrice >> costPerUnit >> fixedCost;

    revenue = units * sellingPrice;
    variableCost = units * costPerUnit;
    totalCost = variableCost + fixedCost;
    profit = revenue - totalCost;

    cout << "Total Revenue: Rs. " << revenue << endl;
    cout << "Variable Cost: Rs. " << variableCost << endl;
    cout << "Total Cost: Rs. " << totalCost << endl;
    cout << "Profit: Rs. " << profit << endl;

    return 0;
}

```

Simple explanation: Revenue is calculated first. Then variable cost and fixed cost are added to get total cost. Profit is revenue minus total cost.

Q4. Employee Salary Calculator

Calculate gross salary and net salary by adding allowances and subtracting tax.

```

#include <iostream>
using namespace std;

int main() {
    double basicSalary, houseAllowance;
    double medicalAllowance, transportAllowance, tax;
    double grossSalary, netSalary;

    cin >> basicSalary;
    cin >> houseAllowance;
    cin >> medicalAllowance;
    cin >> transportAllowance;
    cin >> tax;

    grossSalary = basicSalary + houseAllowance
                  + medicalAllowance + transportAllowance;
    netSalary = grossSalary - tax;

    cout << "Gross Salary: Rs. " << grossSalary << endl;
    cout << "Net Salary: Rs. " << netSalary << endl;

    return 0;
}

```

Simple explanation: All allowances are added to basic salary. Tax is then subtracted from gross salary.

Q5. Discount and Final Price

Take price, quantity and discount percentage and calculate the final amount.

```
#include <iostream>
using namespace std;

int main() {
    double price, discountPercent;
    int quantity;
    double grossAmount, discountAmount, finalAmount;

    cin >> price >> quantity >> discountPercent;

    grossAmount = price * quantity;
    discountAmount = grossAmount * (discountPercent / 100);
    finalAmount = grossAmount - discountAmount;

    cout << "Gross Amount: Rs. " << grossAmount << endl;
    cout << "Discount: Rs. " << discountAmount << endl;
    cout << "Final Amount: Rs. " << finalAmount << endl;

    return 0;
}
```

Simple explanation: First I find the total price, then calculate the discount and subtract it from the total.

Q6. Sales Commission Calculator

Calculate commission from total sales and a commission rate.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    double salesAmount, commissionRate, commission;

    cout << "Enter salesperson name: ";
    getline(cin, name);
    cout << "Enter total sales: ";
    cin >> salesAmount;
    cout << "Enter commission rate: ";
    cin >> commissionRate;

    commission = salesAmount * (commissionRate / 100);

    cout << "Salesperson: " << name << endl;
    cout << "Total Sales: Rs. " << salesAmount << endl;
    cout << "Commission: Rs. " << commission << endl;

    return 0;
}
```

Simple explanation: Commission is simply sales multiplied by the percentage rate.

Q7. Break-Even Calculation

Find break-even units using fixed cost, selling price and variable cost per unit.

```
#include <iostream>
using namespace std;

int main() {
    double fixedCost, sellingPrice, variableCost;
    double breakEvenUnits;

    cin >> fixedCost >> sellingPrice >> variableCost;

    breakEvenUnits = fixedCost /
        (sellingPrice - variableCost);

    cout << "Break Even Units: "
        << breakEvenUnits << endl;

    return 0;
}
```

Simple explanation: The formula used is fixed cost divided by (selling price minus variable cost).

Q8. Sales Target Evaluation

Compare actual sales with the target and show whether the target was achieved.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    double target, actual;

    cout << "Enter salesperson name: ";
    getline(cin, name);
    cin >> target >> actual;

    if (actual >= target)
        cout << name << " has Achieved the Target!" << endl;
    else
        cout << name << " has NOT Achieved the Target!" << endl;

    return 0;
}
```

Simple explanation: An if-else statement checks whether actual sales are equal to or greater than the target.

Q9. Customer Discount Eligibility

Give a discount when the purchase amount is high enough.

```
#include <iostream>
using namespace std;

int main() {
    double amount, discountRate, discount, finalAmount;

    cin >> amount;

    if (amount >= 5000)
        discountRate = 10;
    else if (amount >= 2000)
        discountRate = 5;
    else
        discountRate = 0;

    discount = amount * discountRate / 100;
    finalAmount = amount - discount;

    cout << "Purchase Amount: Rs. " << amount << endl;
    cout << "Discount Rate: " << discountRate << "%" << endl;
    cout << "Discount Amount: Rs. " << discount << endl;
    cout << "Final Payable Amount: Rs. " << finalAmount << endl;

    return 0;
}
```

Simple explanation: The if-else-if conditions decide which discount percentage the customer gets.

Q10. Employee Bonus & Performance

Use the employee's score to show performance classification and calculate bonus.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name, classification;
    double basicSalary, score, bonus, finalSalary;

    cin >> name >> basicSalary >> score;

    if (score >= 90) {
        classification = "Excellent";
        bonus = basicSalary * 0.20;
    } else if (score >= 75) {
        classification = "Good";
        bonus = basicSalary * 0.10;
    } else if (score >= 60) {
        classification = "Average";
        bonus = basicSalary * 0.05;
    } else {
        classification = "Needs Improvement";
        bonus = 0;
    }

    finalSalary = basicSalary + bonus;

    cout << "Name: " << name << endl;
    cout << "Classification: " << classification << endl;
    cout << "Bonus: Rs. " << bonus << endl;
    cout << "Final Salary: Rs. " << finalSalary << endl;

    return 0;
}
```

Simple explanation: The score is checked in different ranges. Each range has its own bonus percentage.

Q11. Simple Billing System

Calculate subtotal, discount, tax and final bill for an item.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string item;
    int quantity;
    double price, subtotal, discount, tax, total;

    cin >> item >> price >> quantity;

    subtotal = price * quantity;
    discount = subtotal * 0.10;
    tax = (subtotal - discount) * 0.05;
    total = subtotal - discount + tax;

    cout << "Item: " << item << endl;
    cout << "Quantity: " << quantity << endl;
    cout << "Subtotal: Rs. " << subtotal << endl;
    cout << "Discount: Rs. " << discount << endl;
    cout << "Tax: Rs. " << tax << endl;
    cout << "Total Amount: Rs. " << total << endl;

    return 0;
}
```

Simple explanation: The bill is found step by step: subtotal, discount, tax and then the final amount.

Submitted by: Faiza Bibi (7), Noor Ul Ain (17), Syeda Arzoo (30)

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